Ultrafast Electronic Dynamics in Quantum Materials Probed by Time-Resolved Angle-Resolved Photoemission Spectroscopy

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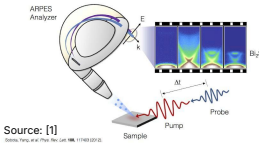
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Introduction

Time-resolved angle-resolved photoemission spectroscopy (**tr-ARPES**) is a technique used to probe ultrafast electron dynamics in advanced materials.

In tr-ARPES, the lower energy **pump pulse** causes initial excitations within the sample, and shortly after the high energy **probe pulse** causes photoemission of electrons. By scanning the pump-probe delay and measuring electrons' kinetic energy and emission angle, the real-time evolution of the electronic band structure can be tracked.



Source: [1]

Figure 1: Diagram of tr-ARPES process for band structure measurements.

Experimental Setup

Our tr-ARPES system uses *Carbide* laser with a central wavelength of 1030 nm. After beam splitting, the frequency-doubled beam drives XUV pulse generation and serves as the probe. The pump path beam is switchable between **515nm** and **1030nm** in our new setup.

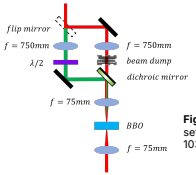


Figure 2: Updated setup for switchable 515nm and 1030nm pump excitation.

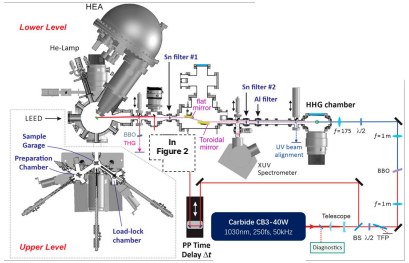


Figure 3: Diagram of the HHG-based tr-ARPES setup, showing the relative paths of the pump (in red) and probe (in blue and pink) pulses. Source: [2]

Methods

Optical Alignment and System Setup:
- The spot size of both the pump and probe were measured; spatial + temporal overlap was performed.
- Optical power was measured at key points, and individual components were adjusted to maximize throughput and confirm proper alignment.

Sample Preparation and Loading:
- TaS_2 and Bi_2Se_3 samples were mounted using a two-component epoxy (1:1).
- Samples transferred from load lock chamber into the main measurement chamber under ultra-high.

Data Analysis:
- Results generated from scans are analyzed and results are plotted using **PyARPES**, an open-source Python framework.

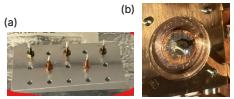


Figure 4: Images of samples (a) after being prepared and (b) loaded into the main chamber.

Time Evolution of TaS_2

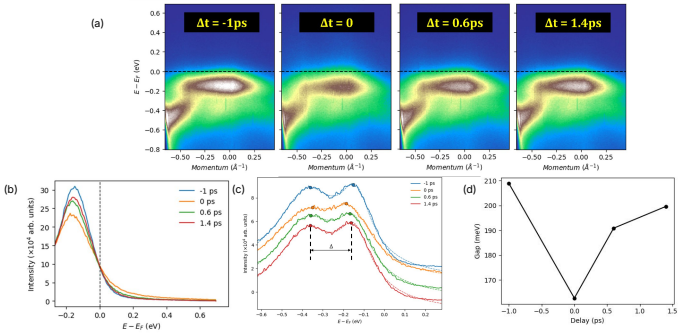


Figure 5: Time-resolved ARPES measurements of 1T-TaS_2, a charge density wave (CDW) material. (a) Evolution of the band structure across various pump-probe delay times, (b) energy distribution curve integrated around Γ, (c) energy distribution curve integrated over the momentum range (-0.34, -0.32)Å^{-1}, (d) CDW gap values corresponding to panel (c).

Single Shot for TaS_2

Under single-shot excitation, TaS_2 can be driven into a metastable state. Increasing fluence unexpectedly produced ring- and fringe-like patterns. The region with the highest beam intensity appears to be the least affected.

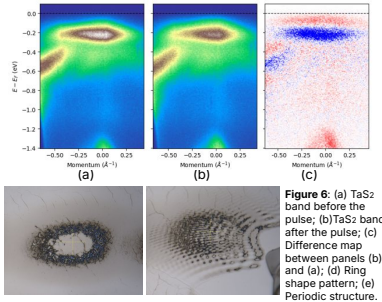


Figure 6: (a) TaS_2 band before the pulse; (b)TaS_2 band after the pulse; (c) Difference map between panels (b) and (a); (d) Ring shape pattern; (e) Periodic structure.

Bi_2Se_3 Measurements

Bi_2Se_3 is a topological insulator material, meaning that it behaves as an insulator on the interior but has conducting states on the surface. Due to this, the band structure of Bi_2Se_3 possesses a signature Dirac cone shape, which can be visualized using ARPES.

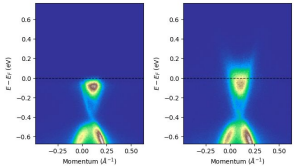
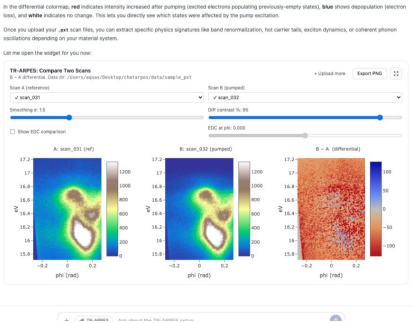


Figure 7: Band structure of Bi_2Se_3.

ChatARPES

Drag-drop a .pxt, get a publication-ready pump-probe differential and a literature-grounded answer in the same window.



TaS_2 pump-probe differential. Red = pump-induced population, blue = depopulation. Gaussian smoothing σ = 1.5, computed client-side.

Conversational Layer
- Claude Haiku 4.5
- Lab papers cached
- Inline citations
- Markdown + LaTeX

Analysis Layer
- Igor .pxt loader
- Two-scan pump-probe
- Plotly heat maps + EDC
- Publication PNG export

Architecture

Researcher → React UI → FastAPI → Haiku 4.5 → .pxt + PDFs

Modular: swap the LLM, point at another lab's PDFs, reuse frontend & engine unchanged.

References & Acknowledgements

[1] Sobota, Yang, *et al*, ***Phys. Rev. Let. 108***, 117403 (2012).
[2] Buss, J. H. *et al*, ***Rev. Sci. Instrum.*** **90**, 023105 (2019).

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